Face Recognition in a Pakistani Context

Team Mamdani · NEDUET CS&IT

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The Brief

- Build an end-to-end face recognition system for Pakistani deployment
- **Collect your own data and fine-tune** a model
- Report accuracy, precision, recall, F1, ROC, FAR/FRR, EER, ablations
- **Per-group fairness** — head coverings, gender, occlusion
- **Anti-spoofing** is required, not optional

Our reading: the hard part is not the embedder. It is detection, calibration, and being honest about what the benchmark can and cannot support.

Headline

98.63%	0.69%	0.9995	0%
rank-1 · 431/437	EER	AUC	failure-to-enrol

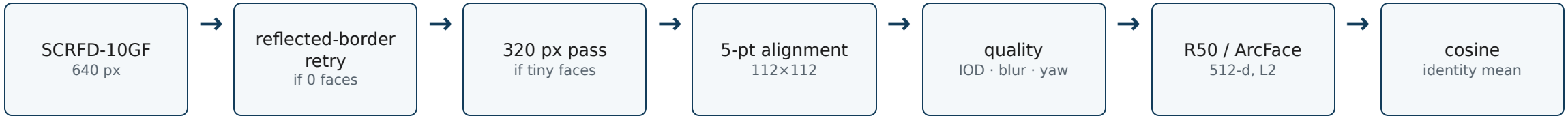
- 23 identities · 849 gallery images · 437 probes · all 437 enrolled
- TAR@1e-3 **97.71%** · precision 0.904 · recall 0.993 · F1 0.947 @ 0.40
- The shipped package, run end to end at threshold 0.40: **430/437 = 98.40%**

Data

Dataset	Role	Size
Pakistani Public Figures	evaluation	23 IDs · 1,286 images (849 gallery / 437 probes)
Self-collected corpus	fine-tune	99 → **98 IDs · 2,230 crops** (54 women, 24 veiled)
LFW	fine-tune ablation	5,749 IDs · 13,233 images
Masked-LFW	occlusion control	200 IDs · 400 clean + 400 masked probes
Niqab sets (Kaggle, OFD)	enrolment only	532 + 400 images, **no identity labels**
NUAA	liveness	150 real + 150 print attacks

No identity overlap between training and evaluation — checked by name before download and again before training. Near-duplicate audit: 2/437 probes; removing both gives 98.62%, so the headline is not leakage-driven.

Pipeline



- **Liveness gate** (MiniFASNet v2 + v1se) sits in front of the matcher when enabled
- Quality failures are **flagged and salvaged**, never silently dropped
- One row out per image in, including `unknown` / `read_error`

The Decisive Finding: Detection, Not the Model

Stage	Probes enrolled	Rank-1	TAR@1e-3
Native SCRFD only	328 / 437 (75.1%)	73.0%	1.14
+ reflected-border retry	**437 / 437 (100%)**	97.94%	97.03
+ 320 px coarse pass	437 / 437	**98.63%**	**97.71**

- 109/437 were tight headshots where SCRFD found **zero** faces; the retry rescued **109/109**
- The 320 px pass fixed **2 silent identity swaps**: a background face was returned instead of the subject
- Swapping R50 for a 4× larger model moved the score by **one probe**

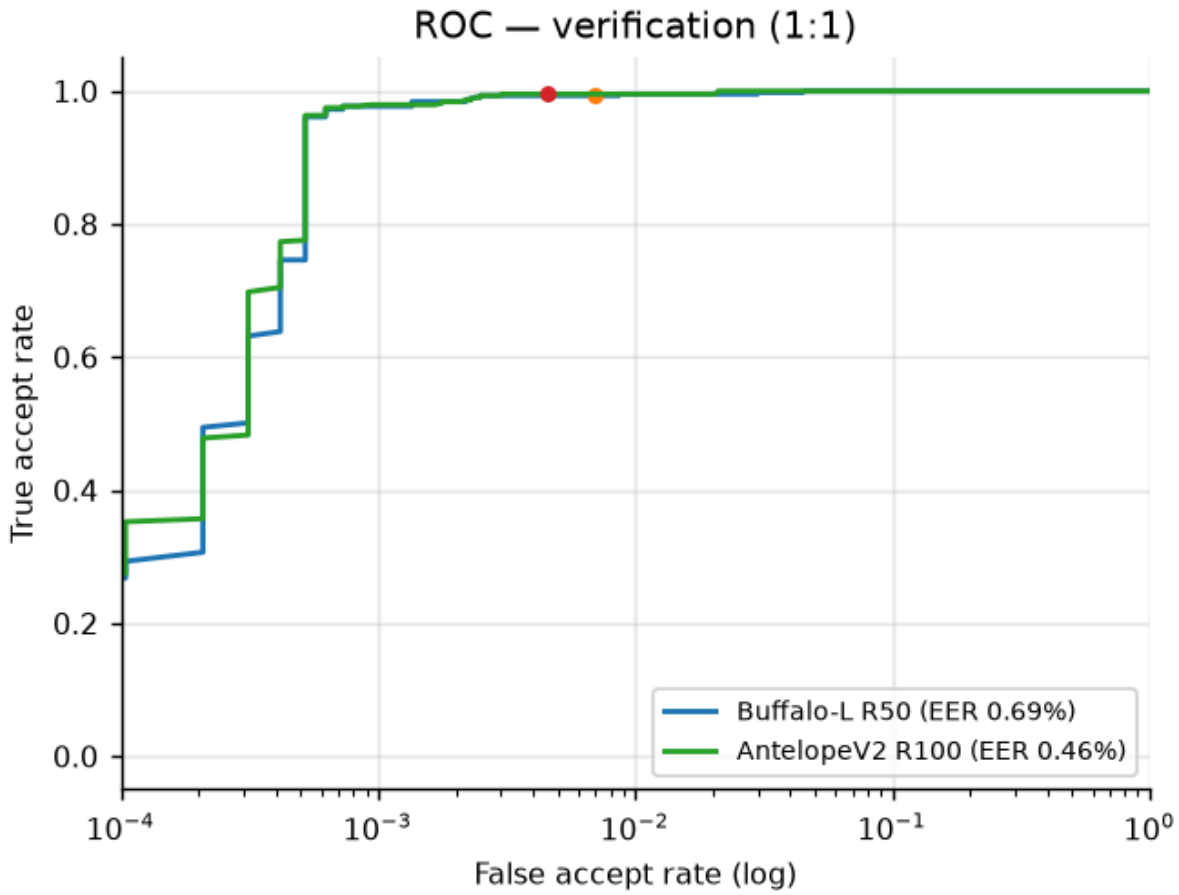
Four Embedders, One Result

Model	Rank-1	TAR@1e-3	EER
buffalo_l R50	98.63%	97.71	0.69%
antelopev2 R100	98.86%	97.94	0.46%
AdaFace IR-101	98.86%	**98.17**	0.46%
AdaFace IR-50	98.63%	97.71	0.46%
Mean fusion of all four	98.86%	97.94	0.46%

- All six errors are **nested**: no model succeeds where another fails
- Fusion cannot help when the mistakes are identical
- **Ship R50:** ties on rank-1, 184 MB of the 435 MB bundle, ~3× faster on CPU
- IR-101 is the better *verification* choice: best genuine–impostor separation

The Operating Point

- Threshold 0.40: **FRR 0.69% · FAR 0.48%**
- 0.35 lowers FRR to 0.46% but doubles FAR to 0.89%
- Shipped CLI gives **98.40%** at 0.40; the harness gives 98.17% — one probe apart for a documented quality-clip artefact, not a different operating point



Veiled Faces: Ranking Survives, Calibration Does Not

Condition	Rank-1	TAR@1e-3	Genuine mean
Clean	97.94%	97.03	0.79
Hijab (hair + sides)	**97.94%**	96.57	0.76
Niqab light (eyes clear)	96.57%	90.39	0.50
Niqab full	91.99%	83.98	0.36
Niqab extreme (eye slit)	89.02%	72.99	0.28

- **The hijab is free** — the five alignment landmarks are untouched
- Even extreme covering keeps the right identity first 89% of the time
- But the score distribution moves: at a fixed 0.40 threshold, niqab probes are returned as `unknown`. Correct ranking, wrong calibration.

Real Niqab Data: The Bottleneck Is Size

Dataset	Images	Detected	Strict quality	Liveness
Kaggle niqab photos	532	**82.1%**	22.0%	**97.9%**
OFD occluded faces	400	61.3%	19.8%	93.9%

- Median inter-ocular distance on detected faces: **48–50 px**, below our 64 px gate
- A 112×112 template from a 50 px face is mostly interpolation
- **Anti-spoofing is not biased against veils** — 94–98% pass, comparable to unveiled
- Real niqab data has no identity labels, so we measure enrolment, never recognition accuracy

Fairness: Report What the Data Can Support

- **Gender, per identity** (not per probe): 20 male IDs → 98.41% (378 probes); 3 female IDs → 100% (59 probes)
- Three identities is **low power** — this is a check, not a fairness claim
- An attribute model labelled 13/23 identities female, including bearded men — which is why per-probe gender labels were rejected
- **Age and skin tone are declared unreportable** rather than estimated
- Hijab subgroup: 97.94% — no measurable penalty for covering hair

Liveness: Included and Measured

- MiniFASNet **v2 + v1se** ensemble, raw BGR 0–255 input, 80×80
- NUAA print attacks, held-out preprocessing slice:

Threshold	APCER	BPCER	BPCER on our 437 probes
0.02	**1.33%**	**1.33%**	**2.29%**
0.05	0.67%	4.00%	4.35%

- UI gate at 0.05: **412/437 = 94.28%** end to end
- Among accepted probes: **98.56%** — the loss is rejection, not misidentification
- Trade-off stated, not hidden: lower the gate and admit more spoofs

The Compliance Fine-Tune: Run, Rejected

- Corpus: **98 identities · 2,230 aligned crops** (54 women, 24 veiled), synthetic hijab/niqab/mask painted on the aligned crop
- AdaFace **IR-50 MS1MV2** + ArcFace head, frozen BN, **40 epochs** on one T4 (2,034 train / 196 held out, ~11 s/epoch)
- Held-out fit check: 100% — the overfitting warning, not a result

Model	Rank-1	TAR@1e-3	EER
buffalo_l R50 (shipped)	**98.63%**	**97.71**	0.69%
IR-50 + corpus fine-tune	98.17%	95.88	1.37%

Rejected on a pre-registered rule (no loss on rank-1 *or* TAR@1e-3). Both lost probes are low-quality captures (22/24 vs 24/24); every other quality bucket is unchanged. 2,230 clean crops cannot teach what MS1MV2 already knows about blur and small faces.

What Failed, and What We’d Do Next

- **LFW fine-tune:** 41.7% vs 97.9% — rejected by the same rule
- **Score fusion:** 98.86%, identical to the best single model
- **Larger embedders:** statistically tied (one probe, McNemar p = 1.0)
- **Upscaling tight crops:** rescued nothing; the reflected border did
- **Masked faces:** ranking holds, threshold does not — recalibrate per condition

Next: per-condition score normalisation, more real veiled data with labels, and a fix for the harness quality-clip artefact before the next benchmark.

What Ships

The package

- `submit/predict.py` — 1:N nested, flat + index, 1:1 pairs
- Vendored ONNX weights, offline, no downloads
- One output row per input, always
- `selftest.py` → **SELFTEST PASSED**
- 435 MB bundle; R50 default; ~0.9 s/image on 8 CPU threads

The demo

```
uv run --extra demo python app/demo.py
```

- Identify · Verify 1:1 · Enrol
- Shows the aligned crop, top-5 scores, liveness, quality and detection trace
- Same `inference.py` as the benchmark

Takeaway

> **Detection got us to 98.6%. The next problem is calibration, not capacity.**
>
> **We report the negative results, the low-power groups, and the one-probe differences — because a benchmark you cannot interrogate is not evidence.**

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